
A Tutorial on the Design and Simulation of Superconducting Quantum Devices using Keysight ADS QuantumPro

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ABSTRACT

This tutorial is a practical guide for researchers designing and simulating superconducting quantum devices in Keysight ADS QuantumPro. Starting from the circuit-QED design cycle, we cover technology setup and layout construction with the Quantum Artwork library, CPW resonator sizing from the technology stackup, full-wave electromagnetic simulation (Method of Moments and Finite Element Method), S-parameter interpretation, energy-participation (EPR) analysis, and quantum-parameter extraction (anharmonicity, dispersive shift, and cross-Kerr). A flip-chip worked example illustrates multilayer technology setup, chip-to-chip coupling, and bump stitching. The material synthesizes concepts and workflows from Keysight QuantumPro application notes and is intended as filler guidance that can be adapted to laboratory-specific process design kits.

Keywords superconducting qubits · QuantumPro · ADS · transmon · CPW resonators · EPR · flip-chip

1 Introduction

Superconducting qubits operate as artificial atoms at milli-kelvin temperatures and gigahertz microwave frequencies. The essential nonlinear element is the Josephson junction: two superconducting electrodes separated by a thin insulator. In the low-excitation (quantum) regime the junction behaves as a nonlinear inductor that, together with shunt capacitance, yields anharmonic energy levels suitable for isolating a computational two-level system.

A typical chip comprises a network of qubits (often cross-shaped or pocket transmons) and coplanar-waveguide (CPW) resonators used for dispersive readout and for mediating qubit-qubit coupling. When the qubit-resonator detuning Δ is large compared with the coupling strength g , the system is in the *dispersive regime*: the resonator frequency shifts by an amount set by the qubit state, enabling readout without destroying the computational subspace.

The PathWave ADS / QuantumPro design cycle is iterative:

1. Build a schematic from parametric Quantum Artwork cells and generate the synchronized layout.

2. Assign a Quantum technology stackup (single-layer or multilayer / flip-chip) and place ports across junctions and launchpads.
3. Run a linear electromagnetic (EM) analysis in which Josephson junctions are replaced by abstract lumped inductances expected at ultra-low drive.
4. Extract classical S-parameters and quantum figures of merit (anharmonicity, cross-Kerr / dispersive shift).
5. Retune geometry or junction inductance until modes are resolved and target parameters are met; export GDS for fabrication.

Planar circuits benefit from Method-of-Moments (MoM) solvers that solve for surface currents rather than volumetric fields, reducing cost for large chips. Finite-element (FEM) eigenmode analysis remains the gold standard for energy-participation quantization when a full Kerr matrix is required.

This tutorial walks through each stage with the equations and tables a new QuantumPro user needs to size devices, set up simulations, and interpret extracted quantum parameters.

2 Layout Design

2.1 Defining Layers

After creating a workspace, select a Quantum technology library. In recent ADS releases this appears as **Quantum Technology Library** with options such as **Single-Layer**, **Multi-Layer** (flip-chip), and **Coaxmon**. Completing the technology wizard creates a `tech:subst` stackup under the workspace.

A representative single-layer stackup for cryogenic aluminum-on-silicon designs is summarized in Table 1. Conductors are often treated as perfect electric conductors (PEC) to mimic superconducting metals at millikelvin temperatures; silicon is assigned a cryogenic permittivity $\epsilon_r \approx 11.5$ and a very small loss tangent (e.g. 10^{-5}).

Layer	Role	Typical material	Notes
Boundary	Chip outline / outer ground extent	—	Rectangle enclosing all artwork
Conductor (cond)	Signal metal: CPW centers, qubit islands, pads	Al / Nb (PEC in EM)	Primary signal layer
Keep-out	Gap between conductor and ground fill	—	Sets CPW gap / island clearance
Ground-fill	Ground plane between boundary and keep-out	Al / Nb (PEC)	Return path for CPW modes
Airbridge conductor / via	Ground stitching over CPW centers	Metal + via	Suppresses unwanted slotline / odd mode
Inductor (abstract)	Placeholder for Josephson junction	Lumped LC	Simulation / layout placeholder; not fabricated
Air / dielectric	Overburden and substrate	Air, Si, etc.	Edited in stackup / CILD

Table 1: Common Quantum technology layers used when exporting layouts and GDS blueprints.

Open `tech:subst` to edit thicknesses and material assignments. For CPW impedance and resonator sizing, use the **Controlled Impedance Line Designer** (CILD) from the technology tools menu (the “ $Z = 50$ ” icon). Configure a coplanar single-ended line (e.g. Top Plane `<None>`, Signal `cond`, Bottom Plane `<None>` for an open air-backed CPW) and set width W and clearance (gap) g .

Running CILD reports transmission-line properties and the per-unit-length RGLC parameters. For a cryogenic silicon CPW with $(g, W, g) = (9, 15, 9) \sim \mu m$ at $\sim 10 \sim GHz$, a typical result is

$$Z_0 \approx 50.5\Omega, \quad \varepsilon_{\text{eff}} \approx 6.15, \quad L' \approx 418 \frac{\text{nH}}{m}, \quad C' \approx 164 \frac{\text{pF}}{m}. \quad (1)$$

These numbers feed resonator length calculations (Equation 2) before any full-wave run.

2.2 Creating Components

QuantumPro ships a parametric Quantum Artwork library. Core cells used in planar chips include:

Cell	Typical use
Q_TransmonCross / Q_TransmonPocket	Transmon qubit islands
Q_CpwLine, Q_CpwMeander, Q_CpwCurvedBend	Feedlines, buses, readout resonators
Q_CpwCouplerT	Capacitive tee coupler to a feedline
Q_CpwLaunchpadWirebond	Wirebond pads / EM excitation ports
Q_CpwLineOpen	Open stub termination
Q_InductorAbstract	Lumped JJ model for linear EM / quantum extraction
Q_CapacitorNInterdigital	Feed coupling capacitors
Q_Bump (multilayer)	Signal transition between chips

Table 2: Selected Quantum Artwork components for superconducting circuit layout.

In the schematic, set orientation, length, width, gap, and connector options; the layout regenerates automatically. Meander resonators (**Q_CpwMeander**) preserve user-defined start/end offsets while adjusting meander count to meet a target electrical length. CPW cells can include periodic airbridges that short the coplanar grounds across the center conductor, suppressing the antisymmetric “odd” mode that degrades coherence and readout contrast.

Josephson junctions in layout. For EM, place pins across the qubit gap and insert **Q_InductorAbstract** with inductance L_J (and optional junction capacitance C_J). Fabrication GDS typically keeps a dedicated inductor layer as a **placeholder** for the real junction process; that layer is not metalized in the foundry mask set.

Sizing a quarter-wave readout resonator. With effective permittivity from CILD, the physical length of an open (or shorted) quarter-wave resonator targeting frequency f is

$$\ell = \frac{\lambda}{4} = \frac{c}{4f\sqrt{\epsilon_{\text{eff}}}}, \quad (2)$$

where c is the vacuum speed of light. For $f = 6 \sim \text{GHz}$ and $\epsilon_{\text{eff}} = 6.1453$,

$$\ell = \frac{3 \times 10^8}{4 \times 6 \times 10^9 \sqrt{6.1453}} \approx 5042 \sim \mu\text{m}. \quad (3)$$

Total distributed capacitance and inductance follow as $C_{\text{tot}} = C'\ell$ and $L_{\text{tot}} = L'\ell$. Consistency check using the distributed resonance condition:

$$f = \frac{1}{4\sqrt{C_{\text{tot}}L_{\text{tot}}}}. \quad (4)$$

Caveat on lumped formulas. The familiar $f = \frac{1}{2\pi\sqrt{L_{\text{res}}C_{\text{res}}}}$ applies to an **equivalent lumped** RLC model near resonance, not to the raw distributed totals. Equating the shorted quarter-wave input impedance to a parallel RLC near ω_0 yields Equation 5 and Equation 6:

$$C_{\text{res}} = \frac{\pi}{4\omega_0 Z_0} = \frac{C_{\text{tot}}}{2}, \quad (5)$$

$$L_{\text{res}} = \frac{1}{\omega_0^2 C_{\text{res}}} = \left(\frac{8}{\pi^2}\right) L_{\text{tot}}. \quad (6)$$

Use these equivalents when mapping a CPW resonator onto a lumped black-box model.

3 Electromagnetic Simulation

Because the Josephson junction is strongly nonlinear, QuantumPro’s linear EM flow replaces it with the small-signal inductance expected in the quantum (ultra-low power) regime. Enabling vs.

disabling this inductance corresponds to the quantum ground-state vs.

classical high-power regimes: resonator peaks shift slightly when the qubit is “turned on,” and that shift is the dispersive information used for parameter extraction.

3.1 Simulation Setup

A minimal setup for a feedline-coupled qubit–resonator circuit is:

1. **Ports.** Place input/output ports on launchpads (signal to ground). Place a lumped port across each qubit junction where `Q_InductorAbstract` attaches.
2. **Frequency span.** Sweep a band that covers all qubit and resonator modes (often 4–10 ~ GHz for GHz-class devices). Add dense local sweeps near expected resonances when quality factors are high.
3. **Solver choice.** Momentum RF (quasi-static Green functions) for fast early design; Momentum Microwave or FEM for higher accuracy at microwave frequencies; FEM eigenmode for Energy Participation Analysis.
4. **Mesh.** Increase cells-per-wavelength (cpw) globally and refine near qubit islands and junction ports. Keep ground planes coarser. Example starting point for MoM: 100–200 ~ cpw globally, 1000–2000 ~cpw near qubit ports; for FEM EPR, ~ 5 ~ μm edge length on center conductors and islands.
5. **Materials.** Confirm cryogenic ε_r , loss tangent, and PEC conductors in the stackup.

Table 3 compares the QuantumPro analysis paths.

Analysis	Solver	Primary outputs	Best for
Full EM (QS)	Momentum RF	S-parameters, $C(f)$, χ for neighbors	Fast iteration
Full EM (FD)	Momentum Microwave / FEM	S-parameters, dressed modes	High-frequency accuracy
Energy Participation	FEM eigenmode	Eigenfrequencies, Q , T_1 , full χ matrix	Complete matrix, Kerr highest fidelity

Table 3: QuantumPro analysis modes for superconducting circuits.

3.2 Finite Element Method

The FEM discretizes Maxwell’s equations over a volumetric mesh (tetrahedra / hexahedra) subject to the boundary conditions of the computational domain. Relative to MoM—which solves integral equations for surface currents on conductors—FEM:

- Captures full three-dimensional field structure, including substrate modes, chip-to-chip cavities, and radiation into air boxes.
- Supports **eigenmode** solutions: natural frequencies ω_m , quality factors Q_m , and modal field patterns without a driven port sweep.

- Scales with the number of volumetric unknowns and is typically more expensive than MoM for large planar chips.

In QuantumPro, FEM is the engine behind Energy Participation Analysis. Meshing strategy remains decisive: refine conductors that store electric energy at qubit frequencies (charge islands, CPW centers, coupling pads) while coarsening bulk ground and distant air regions. For flip-chip devices, also refine bump interfaces and overlapping coupling patches.

MoM and FEM frequency predictions for the same geometry often agree within a few percent; MoM can slightly underestimate capacitance (hence slightly higher resonant frequencies) unless islands are finely meshed.

3.3 S-Parameters

Driven Full EM Analysis returns the scattering matrix $\mathbf{S}(\omega)$ among defined ports. For a two-port feedline, the transmission S_{21} shows peaks (or dips, depending on coupling topology) at resonator and, when junction inductances are present, qubit resonances. Design goals typically include:

- Distinct, well-separated peaks for every qubit and every readout / bus resonator.
- Target frequencies for each mode (set by L_J , island capacitance, and resonator length).
- Adequate coupling for readout without excessive Purcell decay.

Admittance at a qubit port is obtained from the port S-parameters. Resonance of qubit n (with junction inductance $L_{j,qn}$ and optional $C_{j,qn}$) is located by the zeros of Equation 7:

$$f_{Pn'}(\omega) = \Im[Y_{Pn'Pn'}(\omega)] - \frac{1}{\omega L_{j,qn}} + \omega C_{j,qn}, \quad (7)$$

where $Y_{Pn'Pn'}$ is the admittance seen at the junction port **without** the junction LC. The function $\Im[Y]$ alone zeros at the **classical** (high-power) resonator frequencies ω_r ; including the junction LC moves those zeros to the **dressed** frequencies $\omega_{r'}$. The difference is the dispersive shift

$$\text{D.S.} = \frac{g^2}{\Delta}. \quad (8)$$

In continuous-wave cryogenic measurements the same classical vs. dressed spectra appear when the drive power is swept from high to low on a network analyzer.

3.4 Boundary Conditions

Correct boundary conditions are as important as mesh density:

- **PEC conductors.** Superconducting films are commonly modeled as perfect conductors in the linear EM flow (optionally with kinetic inductance via specialized stackup examples).
- **Ground stitching.** Airbridges and, in flip-chip, bump arrays connect coplanar grounds so that the desired CPW even mode is enforced.
- **Radiation / absorbing boundaries.** FEM domains require a finite air box; absorbing or radiation boundaries prevent artificial cavity modes from contaminating the spectrum.
- **Symmetry and chip outline.** The boundary layer defines the extent of the ground plane; incomplete grounds create spurious slotline modes.
- **Port calibration.** Lumped ports across junctions must span the gap cleanly; launchpad ports should reference the local ground.

When peaks fail to resolve, first check for overlapping modes of **like** elements (qubit–qubit or resonator–resonator), incomplete grounds, missing airbridges, and under-resolved meshes near islands—before changing target frequencies.

3.5 Energy Participation Analysis

Energy Participation Ratio (EPR) quantization starts from modal field solutions. For any resonant mode the time-averaged inductive and capacitive energies are equal. In a Josephson circuit the inductive energy includes kinetic inductance of the junctions:

$$E_{\text{ind}} = E_{\text{cap}} = E_{\text{elec}} = E_{\text{mag}} + E_{\text{kin}}. \quad (9)$$

The participation of junction j in mode m is

$$p_{mj} = \frac{E_{\text{kin}}^{(m,j)}}{E_{\text{ind}}^{(m)}} = \frac{E_{\text{elec}}^{(m)} - E_{\text{mag}}^{(m)}}{E_{\text{elec}}^{(m)}}, \quad (10)$$

or, equivalently from junction current,

$$p_{mj} = \left(\frac{1}{2}\right) L_j I_{m,j}^2 / E_{\text{ind}}^{(m)}. \quad (11)$$

QuantumPro obtains $I_{m,j}$ from the line integral of the electric field across the junction path (voltage drop) divided by $j\omega L_j$.

The Kerr matrix elements follow from the participations and junction energies $E_j = \frac{\Phi_0^2}{4\pi^2 L_j}$:

$$\chi_{mn} = \sum_{j=1}^J \frac{\hbar\omega_m\omega_n}{4E_j} p_{mj}p_{nj} \quad (m \neq n), \quad (12)$$

$$\alpha_{q/r} = \chi_{mm} = \sum_{j=1}^J \frac{\hbar\omega_m^2}{8E_j} p_{mj}^2. \quad (13)$$

Unlike frequency-domain Full EM extraction—which typically returns cross-Kerr only between **directly coupled** neighbors—EPR yields a dense χ matrix including weak couplings between non-adjacent modes (often kilohertz-scale). Eigenmode results also report quality factors and photon lifetimes T_1 for each mode.

3.6 Quantum Parameter Extraction

Anharmonicity from capacitance. For a transmon, the lowest transition frequencies are

$$\omega_{01} = \sqrt{8E_J E_C} - E_C, \quad \omega_{12} = \sqrt{8E_J E_C} - 2E_C, \quad (14)$$

so the anharmonicity is

$$\alpha = \omega_{12} - \omega_{01} = -E_C, \quad E_C = \frac{e^2}{2C}, \quad (15)$$

with C the total capacitance seen from the qubit port at resonance (extractable from the port reflection / capacitance vs.

frequency in the quasi-static solution). Example: $C \approx 61.5 \sim \text{fF}$ yields $\alpha \approx 315 \sim \text{MHz}$.

Cross-Kerr in the dispersive regime. Between a qubit and a resonator,

$$\chi_{qr} = \frac{2g^2\alpha_q}{\Delta(\Delta + \alpha_q)}, \quad (16)$$

with detuning $\Delta = \omega_q - \omega_r$ (sign convention as in the design notes). Full EM analysis also constructs resonator anharmonicity α_r from effective L_r , C_r seen at the qubit port, then forms

$$\chi_{qr} = 2\sqrt{\alpha_q\alpha_r} \quad (17)$$

with

$$C_r = \frac{1}{2} f'_{Pn}(\omega_r), \quad L_r = \frac{1}{\omega_r^2 C_r}. \quad (18)$$

Table 4 and Table 5 reproduce representative QuantumPro comparisons from Keysight application notes (illustrative reference values for mesh and geometry in those examples).

Qty	QS	FD	EPR	Unit	Notes
C_{Q1}	61.53	61.53	—	fF	From port capacitance
α_{Q1}	314.82	314.84	298.80	MHz	Within $\sim 5\%$
χ_{Q1R1}	2.14	2.14	2.02	MHz	Neighbor coupling
α_{R1}	3.62	3.63	3.43	kHz	Resonator self-Kerr

Table 4: One-qubit / one-resonator extraction comparison (Momentum RF quasi-static, Momentum Microwave frequency-domain, FEM EPR). Values adapted from Keysight Quantum Parameter Extraction notes.

Coupling	χ QS (MHz)	χ FD (MHz)	χ EPR (MHz)
χ_{Q1R2}	1.70	1.71	1.62
χ_{Q1R4}	1.22	1.23	1.09
χ_{Q2R1}	2.06	2.07	1.87
χ_{Q2R4}	1.42	1.43	1.23
χ_{Q3R2}	2.26	2.27	1.82
χ_{Q3R3}	1.69	1.70	1.39
χ_{Q4R1}	2.97	2.99	2.42
χ_{Q4R3}	1.80	1.80	1.51

Table 5: Four-qubit / four-resonator neighbor cross-Kerr strengths across solvers. EPR additionally reports non-adjacent couplings (e.g. $\chi_{Q1R3} \sim 1.2 \sim \text{kHz}$) absent from Full EM tables.

Practical tuning knobs.

- Increase (decrease) L_J to lower (raise) qubit frequency: $\omega_0 \sim 1/\sqrt{LC}$.
- Lengthen (shorten) CPW resonators to lower (raise) f_r .
- Adjust coupler geometry (`SecondParallelLength`, claw length, overlap area) to set g and thus χ .

3.7 Time Dynamics Analysis

Frequency-domain and eigenmode extractions characterize the linearized circuit near the vacuum / few-photon regime. Time-domain and nonlinear circuit analyses complement that picture when researchers need drive-dependent dynamics:

- **Dispersive readout waveforms.** With extracted χ and resonator κ , classical IQ trajectories under a readout pulse can be simulated in a circuit or system-level tool using the effective Hamiltonian $H/\hbar = \omega_r a^\dagger a + (1/2)\omega_q \sigma_z + \chi a^\dagger a \sigma_z$.

- **Harmonic balance / transient with JJ models.** ADS circuit simulation (outside the linear EM quantum-parameter flow) can include nonlinear Josephson elements (JJ2, JJ3) for large-signal behavior, Rabi driving, and higher-level leakage estimates—subject to the usual caveats that full open-system master equations are often handled in dedicated quantum dynamics packages.
- **Ring-down and T_1 .** Eigenmode quality factors from EPR map to photon lifetimes $T_1 \approx Q/\omega$; comparing simulated Q against measured ring-down validates loss models (dielectric participation, seam loss, package modes).
- **Power-dependent spectra.** Sweeping drive amplitude in measurement (or in a nonlinear model) recovers the classical $\rightarrow$ quantum transition used to identify dressed resonator peaks—the experimental counterpart of enabling `Q_InductorAbstract` in Full EM Analysis.

A recommended workflow is: (1) converge geometry and χ matrix in QuantumPro, (2) export black-box parameters into a dynamics model for gate / readout pulse design, (3) close the loop with cryogenic S-parameter and time-domain measurements.

4 Flip-Chip Design Example

Routing congestion grows quickly with qubit count. Flip-chip (face-to-face) integration places circuitry on two chips separated by a few micrometers of vacuum / helium, connected by superconducting bumps. QuantumPro supports this via **Quantum Multi-Layer** technology (two layer sets, one flipped).

Technology setup (sketch). Choose two stacks; set substrate thickness (e.g. $280 \sim \mu m$ Si), metal thickness (often $0 \sim$ for PEC sheets), airbridge dielectric height (e.g. $1 \sim \mu m$), and inter-chip **Top Dielectric** air gap (e.g. $6 \sim \mu m$). With $1 \sim \mu m$ airbridges on each side and $6 \sim \mu m$ gap, the conductor-to-conductor spacing is $\sim 8 \sim \mu m$. Define vias / bumps that galvanically stitch grounds (and optionally signal lines).

Two CPW environments.

- **Type A — metal facing metal:** Top Plane 2_cond, Signal cond. Fields partially divert into the inter-chip air gap $\rightarrow$ lower ϵ_{eff} , longer resonators, strong sensitivity to gap height via image return currents.
- **Type B — metal facing dielectric:** Top Plane <None>. Inductance is more stable vs. gap; capacitance still varies as fields penetrate the opposite silicon.

For Type A with $(g, W, g) = (12, 12, 12) \sim \mu m$ at $6 \sim \text{GHz}$, CILD may report $Z_0 \approx 47 \sim \Omega$, $\epsilon_{\text{eff}} \approx 4.55$, $L' \approx 335 \sim \text{nH/m}$, $C' \approx 151 \sim \text{pF/m}$, giving

$$\ell_{\lambda/4} = \frac{c}{4f\sqrt{\epsilon_{\text{eff}}}} \approx 5859 \sim \mu m \quad (19)$$

— roughly 17% longer than a comparable single-chip resonator with $\epsilon_{\text{eff}} \sim 6.2$, consistent with $\ell_{\text{flip}}/\ell_{\text{single}} = \sqrt{\epsilon_{\text{eff, single}}/\epsilon_{\text{eff, flip}}}$.

Ballpark Xmon capacitance. Approximating four CPW arms of length ℓ_{arm} as

$$C_{\text{qubit}} \approx 4\ell_{\text{arm}}C' \quad (20)$$

with $\ell_{\text{arm}} = 165 \sim \mu m$ and $C' = 151.2 \sim \text{pF/m}$ yields $C_{\text{qubit}} \approx 100 \sim \text{fF}$ and $E_C/h \approx e^2/(2hC) \approx 194 \sim \text{MHz}$ —an underestimate that ignores corner fringing and the central pad, but useful before FEM.

Layout recipe (single qubit + readout).

1. Place feedline, Q_CpwCouplerT, and meander resonator on stack 1 (bottom).
2. Place Q_TransmonCross and Q_InductorAbstract on stack 2 by setting LayerGroupPrefix to "2_".
3. In layout, extend the resonator with cond / keepout paths and an overlap pad (e.g. concentric circles) under the cross to set coupling capacitance.
4. Draw boundary and 2_boundary ground outlines; stitch grounds with Bump_1_2 circles (do **not** use Q_Bump for ground stitching—Q_Bump is for CPW signal transitions and includes keep-out).
5. Open QuantumPro, verify Z-exaggerated geometry, refine mesh on cond, 2_cond, and bumps, and run Full EM and/or EPR.

Illustrative extracted numbers (Momentum Microwave vs.

FEM EPR on a tutorial geometry) appear in Table 6. From EPR anharmonicity one recovers

$$C_{\text{qubit}} = \frac{e^2}{2E_C}, \quad (21)$$

and from cross-Kerr

$$\chi = \frac{g^2 E_C}{\Delta(\Delta - E_C)} \quad (22)$$

one solves for g , then for the geometric coupling capacitance C_g via the standard capacitive-coupling formula linking g to C_g , C_{qubit} , and C_{res} (with $C_{\text{res}} = C_{\text{tot}}/2$ for a quarter-wave resonator).

Quantity	Full EM (MoM)	EPR (FEM)
Qubit frequency	$\sim 4.55 \sim \text{GHz}$	$\sim 4.37 \sim \text{GHz}$
Resonator frequency	$\sim 7.1 \sim \text{GHz}$	$\sim 7.36 \sim \text{GHz}$
Qubit anharmonicity E_C/h	(from χ table)	$\sim 159 \sim \text{MHz}$
Cross-Kerr χ	(from S -param extract)	$\sim 0.17 \sim \text{MHz}$
Implied C_{qubit}	—	$\sim 122 \sim \text{fF}$
Implied g	—	$\sim 96 \sim \text{MHz}$

Table 6: Order-of-magnitude flip-chip qubit–resonator results from a QuantumPro tutorial geometry (values adapted from Keysight flip-chip application notes). Mesh refinement reduces MoM–FEM discrepancies.

Design warning. Image currents on a metal plane above a meander track the resonator shape and **reduce** L' , shifting f_r when the chip gap varies. Mitigation strategies include facing resonators toward dielectric (Type B), etching apertures in the opposite ground to break return paths, or lithographic interruption of image-current loops.

5 Conclusion

Keysight ADS QuantumPro provides an end-to-end path from parametric superconducting layouts to fabrication GDS and to quantum figures of merit extracted from linear EM. Researchers should: size CPWs and resonators with CILD against the real stackup; enforce clean grounds and airbridges; choose MoM for rapid Full EM iteration and FEM EPR when a complete Kerr matrix or highest accuracy is required; and treat flip-chip gap sensitivity as

a first-class design constraint. The equations and tables above are intended as working filler content—replace numerical examples with your process parameters and close the loop against cryogenic measurements.

APPENDIX A References and further reading

This tutorial synthesizes publicly documented Keysight QuantumPro application-note material and standard circuit-QED references. Consult the ADS Help tree under **Applications** → **Quantum** for the latest demos and PDFs shipped with your install.

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